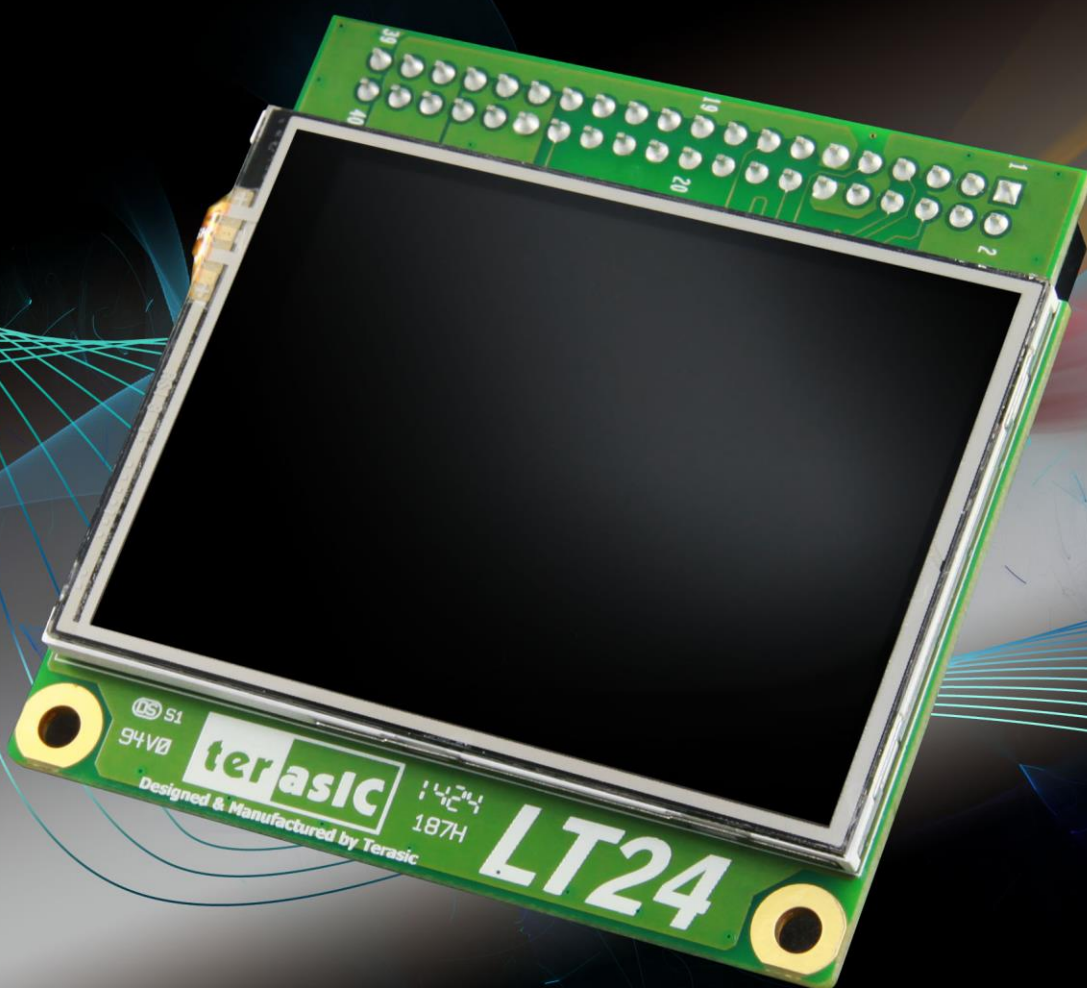


LT24

User Manual



CHAPTER 1	<i>INTRODUCTION</i>	3
1.1	The Package Contents	3
1.2	Assemble LT24 with DE0-Nano	4
1.3	Connectivity	5
1.4	Getting Help	7
CHAPTER 2	<i>ARCHITECTURE OF LT24</i>	8
2.1	Features	8
2.2	Layout and Block Diagram	8
CHAPTER 3	<i>USING THE LT24</i>	10
3.1	Pin Definition of 2x20 GPIO Connector	10
3.2	Using LCD	12
3.3	Using Touch	13
CHAPTER 4	<i>PAINTER DEMONSTRATION</i>	15
4.1	Description	15
4.2	System Description	16
4.3	LT24 LCD Controller	17
4.4	Setup Painter Demo on Terasic DE2-115 FPGA Board	17
4.5	Painter Demo for Other Terasic FPGA Mainboards	18
CHAPTER 5	<i>LT24 DISPLAY</i>	19
5.1	Description	19
5.2	System Description	20
5.3	LT24 LCD Initialization	20
5.4	Setup Demo on Terasic DE0-Nano FPGA Board	21
5.5	LT24_Display Demo for Other Terasic FPGA Mainboards	22
CHAPTER 6	<i>APPENDIX</i>	23



6.1 Revision History	23
6.2 Copyright Statement	23

Chapter 1

Introduction

The Terasic LT24 is a 2.4" LCD touch module with 240(H) x 320(V) display resolution. It can be bundled with various Terasic FPGA development boards through the 2x20 GPIO interface. The LT24 is powered directly from the FPGA mainboard. It doesn't require any power adaptor. The kit contains complete reference design and source code for the Painter application. This chapter provides the key information about the kit.

1.1 The Package Contents

The LT24 kit comes with the following items:

- LT24 touch LCD module
- Two silicon footstands
- Two screw nuts
- Two hexagon copper pillars
- CD download guide

The system CD contains technical documents of LT24 kit, which include component datasheets, demonstrations, schematic and user manual. Users can download the CD from the link below:

<http://cd-lt24.terasic.com>

Figure 1-1 shows the contents of LT24 kit.



Figure 1-1 Contents of LT24 kit

1.2 Assemble LT24 with DE0-Nano

There are two holes reserved on the LT24 for the screws to stabilize the installation with other mainboards by adding copper pillars onto them. The following steps take DE0-Nano as an example to demonstrate the installation with the copper pillars.

1. Install the screw nuts from top to the bottom of screw holes reserved on the LT24. Install the hexagon copper pillars and make sure then are tightened up, as shown in [Figure 1-2](#).

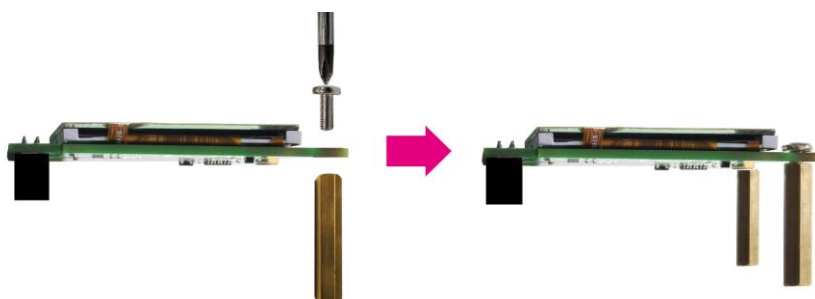


Figure 1-2 Install the hexagon copper pillars

2. Cover the two copper pillars of LT24 with two silicon footstands, as shown in [Figure 1-3](#). This procedure can be skipped if the LT24 is assembled with DE0-Nano.

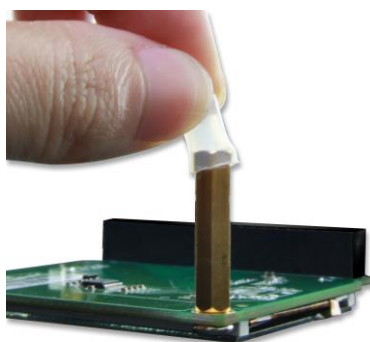


Figure 1-3 Install the footstand

3. Connect the LT24 to the GPIO header of DE0-Nano, as shown in [Figure 1-4](#).

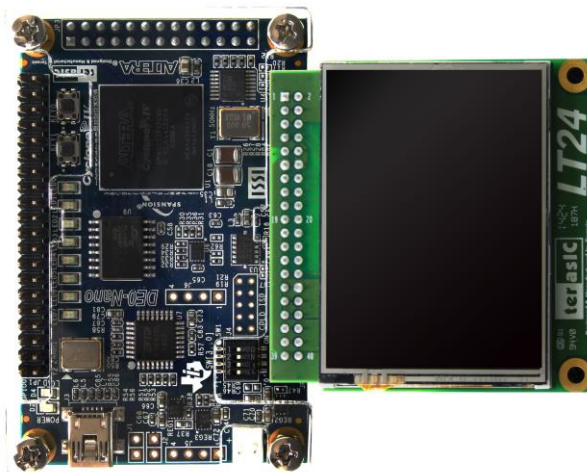


Figure 1-4 LT24 Assembled with DE0-Nano

1.3 Connectivity

Figure 1-5, Figure 1-6, Figure 1-7 and Figure 1-8 below show the connectivity of LT24 to DE0-Nano, DE1-SoC, DE2-115 and C5G(Cyclone V GX Starter Kit), respectively. The LT24 is powered from FPGA mainboard. It is not necessary to connect the LT24 with a power adaptor.

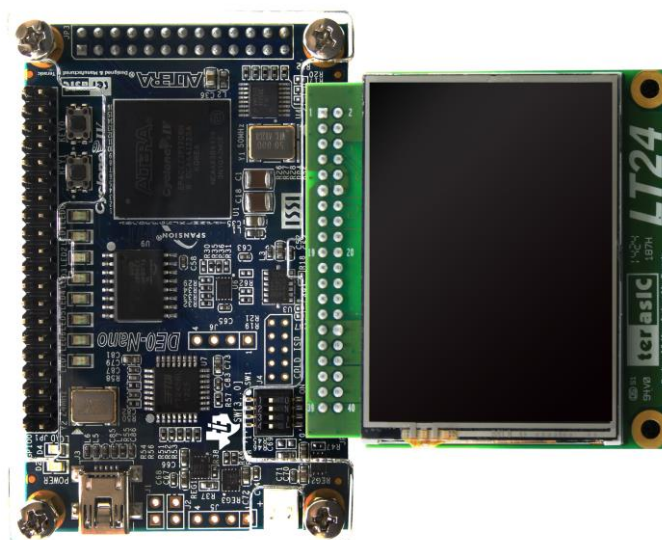


Figure 1-5 LT24 with DE0-Nano

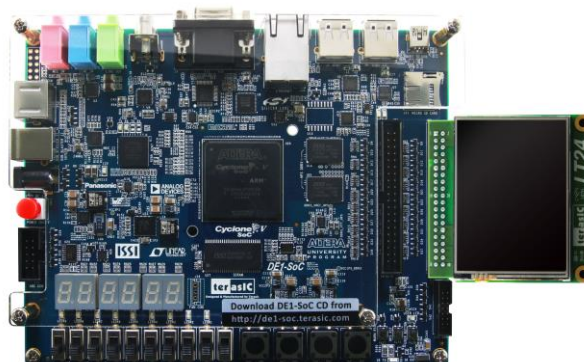


Figure 1-6 LT24 with DE1-SoC

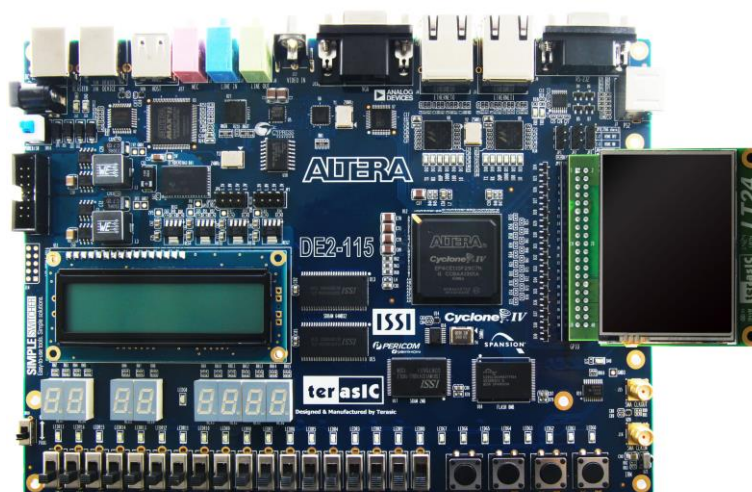


Figure 1-7 LT24 with DE2-115

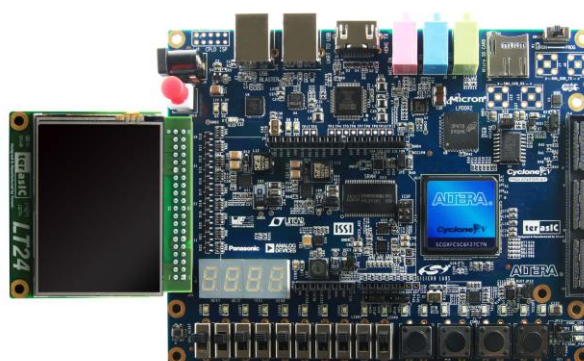


Figure 1-8 LT24 with C5G (Cyclone V GX Starter Kit)

1.4 Getting Help

Here is the information to get help if you encounter any problem:

- Office Hours: 9:00 a.m. to 6:00 p.m. (GMT +8)
- Telephone: +886-3-575-0880
- Email: support@terasic.com

Chapter 2

Architecture of LT24

This chapter lists the features and describes the architecture of 2.4" LCD touch module (LT24).

2.1 Features

The key features of this module are listed below:

- 240(H) x 320 (V) pixel resolution LCD with 65K RGB color
- Single resistive touch
- 2x20 GPIO interface

2.2 Layout and Block Diagram

Component and Layout

The top view of LT24 is shown in **Figure 2-1**.



Figure 2-1 Top view of LT24

The bottom view of LT24 is shown in **Figure 2-2**. It depicts the layout and indicates the locations of connectors and key components.



Figure 2-2 Bottom view of LT24 module

Block Diagram

Figure 2-3 shows the block diagram of LT24. An ADC chip is used to handle the 4-wire touchscreen.

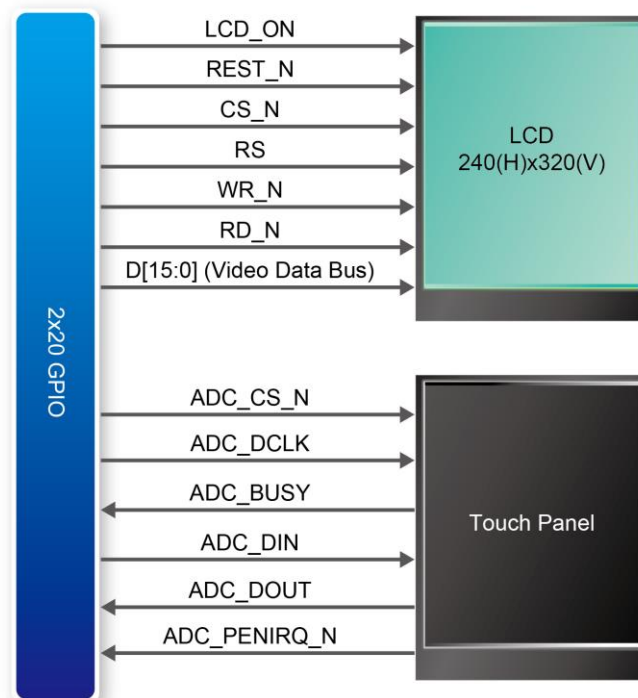


Figure 2-3 Block diagram of LT24

Using the LT24

This chapter provides information on how to control the hardware of 2.4" LCD touch module (LT24), including the definition of 2x20 GPIO interface, LCD control, and touch control signals.

3.1 Pin Definition of 2x20 GPIO Connector

The 2x20 GPIO female connector on LT24 connects directly to the 2x20 GPIO male connector on Terasic FPGA development boards. **Figure 3-1** illustrates the signal names of 2x20 GPIO Header.

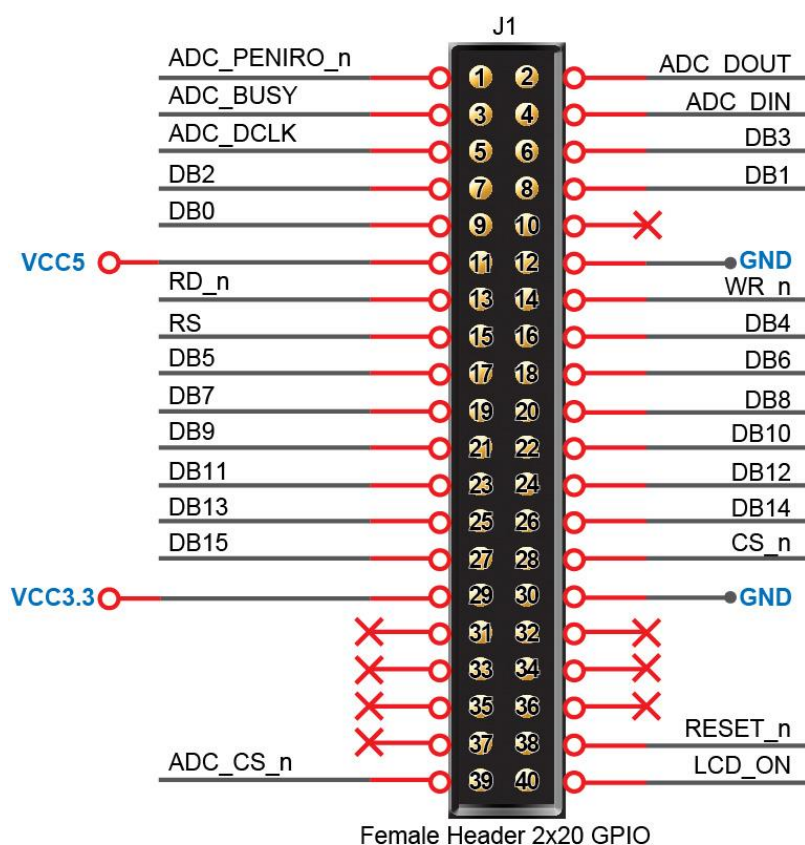


Figure 3-1 Signal names of 2x20 GPIO connector

Table 3-1 shows the LT24 pin assignments for the 2x20 GPIO pins in Quartus II.

Table 3-1 LT24 Pin Assignments of 2x20 GPIO in Quartus II

Pin Numbers	Pin Name	Direction	IO Standard	
1	ADC_PENIRQ_N	Input	3.3-V LVTTL	GPIO[0]
2	ADC_DOUNT	Input	3.3-V LVTTL	GPIO[1]
3	ADC_BUSY	Input	3.3-V LVTTL	GPIO[2]
4	ADC_DIN	Output	3.3-V LVTTL	GPIO[3]
5	ADC_DCLK	Output	3.3-V LVTTL	GPIO[4]
6	D[3]	Output	3.3-V LVTTL	GPIO[5]
7	D[2]	Output	3.3-V LVTTL	GPIO[6]
8	D[1]	Output	3.3-V LVTTL	GPIO[7]
9	D[0]	Output	3.3-V LVTTL	GPIO[8]
10	-	-	-	GPIO[9]
11	-	-	-	N/C
12	-	-	-	N/C
13	RD_N	Output	3.3-V LVTTL	GPIO[10]
14	WR_N	Output	3.3-V LVTTL	GPIO[11]
15	RS	Output	3.3-V LVTTL	GPIO[12]
16	D[4]	Output	3.3-V LVTTL	GPIO[13]
17	D[5]	Output	3.3-V LVTTL	GPIO[14]
18	D[6]	Output	3.3-V LVTTL	GPIO[15]
19	D[7]	Output	3.3-V LVTTL	GPIO[16]
20	D[8]	Output	3.3-V LVTTL	GPIO[17]
21	D[9]	Output	3.3-V LVTTL	GPIO[18]
22	D[10]	Output	3.3-V LVTTL	GPIO[19]
23	D[11]	Output	3.3-V LVTTL	GPIO[20]
24	D[12]	Output	3.3-V LVTTL	GPIO[21]
25	D[13]	Output	3.3-V LVTTL	GPIO[22]
26	D[14]	Output	3.3-V LVTTL	GPIO[23]
27	D[15]	Output	3.3-V LVTTL	GPIO[24]
28	CS_N	Output	3.3-V LVTTL	GPIO[25]
29	-	-	-	N/C
30	-	-	-	N/C
31	-	-	-	GPIO[26]
32	-	-	-	GPIO[27]
33	-	-	-	GPIO[28]
34	-	-	-	GPIO[29]
35	-	-	-	GPIO[30]
36	-	-	-	GPIO[31]
37	-	-	-	GPIO[32]

38	RESET_N	Output	3.3-V LVTTL	GPIO[33]
39	ADC_CS_N	Output	3.3-V LVTTL	GPIO[34]
40	LCD_ON	Output	3.3-V LVTTL	GPIO[35]

3.2 Using LCD

The LCD features 240(H) x 320(V) pixel resolution. The ILI9341 LCD driver is used to drive the LCD display. The 65K-Color, RGB 5-6-5 bits input data based on 8080-system 16-bit parallel bus interface of ILI9341 is used on the LT24. For further information, please refer to the section 7.6.5 “16-bit Parallel MCU Interface” of ILI9341 datasheet in the LT24 System CD. **Figure 3-2** shows the signals connected between the ILI9341 and the FPGA.



Figure 3-2 Signals of LCD driver connected to FPGA

CSX is a low-active chip select pin. **D/CX** is Data or Command selection pin. When **D/CX** = 1, data selected. When **D/CX** = 0, command is selected. **WRX** is a write signal and writes data at the rising edge. **RDX** is a read signal and MCU read data at the rising edge. **D[15:0]** is data bus. **Figure 3-3** shows pixel data transfer format. The D0~D7 are used to translate command to ILI9341 in the command mode (D/CX=0). The D0~D15 are used to translate the RGB data to ILI9341 in data mode (D/CX=1).

Count	0	1	2	3	...	238	239	240
D/CX	0	1	1	1	...	1	1	1
D15		0R4	1R4	2R4	...	237R4	238R4	239R4
D14		0R3	1R3	2R3	...	237R3	238R3	239R3
D13		0R2	1R2	2R2	...	237R2	238R2	239R2
D12		0R1	1R1	2R1	...	237R1	238R1	239R1
D11		0R0	1R0	2R0	...	237R0	238R0	239R0
D10		0G5	1G5	2G5	...	237G5	238G5	239G5
D9		0G4	1G4	2G4	...	237G4	238G4	239G4
D8		0G3	1G3	2G3	...	237G3	238G3	239G3
D7	C7	0G2	1G2	2G2	...	237G2	238G2	239G2
D6	C6	0G1	1G1	2G1	...	237G1	238G1	239G1
D5	C5	0G0	1G0	2G0	...	237G0	238G0	239G0
D4	C4	0B4	1B4	2B4	...	237B4	238B4	239B4
D3	C3	0B3	1B3	2B3	...	237B3	238B3	239B3
D2	C2	0B2	1B2	2B2	...	237B2	238B2	239B2
D1	C1	0B1	1B1	2B1	...	237B1	238B1	239B1
D0	C0	0B0	1B0	2B0	...	237B0	238B0	239B0

Figure 3-3 65K-Color, 5-6-5 bits pixel data format

3.3 Using Touch

The AD7843 from Analog Device is a 12-bit analog-to-digital converter (ADC) used to digitize the touch points from the touch panel of LT24 into the corresponding X/Y coordinates on LT24.

To obtain the coordinates from the ADC, users need to monitor the interrupt signal ADC_PENIRQ_n coming out of the ADC first. The ADC_PENIRQ_n output remains high in normal condition by connecting a pull high resistor to it. When the touch screen connected to the ADC is trigger via a pen or finger, the ADC_PENIRQ_n output goes low and it initiates an interrupt to FPGA, which instructs a control word to be written to the ADC via the serial port interface.

The control word provided to the ADC via the DIN pin is shown in **Table 3-2**. It controls the conversion start, channel addressing, ADC conversion resolution, configuration, and power-down of the ADC. The detailed information about the order and description of these control bits can be found in the datasheet from the DATASHEET folder in the LT24 System CD.

Table 3-2 Control Register Bit Function Description

MSB							LSB
S	A2	A1	A0	MODE	SER/DEF	PD1	PD0
Bit	Mnemonic	Comment					
7	S	Start Bit. The control word starts with the first high bit on DIN. A new control word can start every 15th DCLK cycle when in the 12-bit conversion mode, or every 11th DCLK cycle when in 8-bit conversion mode.					

6-4	A2-A0	Channel Select Bits. These three address bits, along with the $\overline{\text{SER/DEF}}$ bit, control the setting of the multiplexer input, switches, and reference inputs.
3	MODE	12-Bit/8-Bit Conversion Select Bit. This bit controls the resolution of the following conversion. With 0 in this bit, the conversion has a 12-bit resolution, or with 1 in this bit, the conversion has a 8-bit resolution.
2	$\overline{\text{SER/DEF}}$	Single-Ended/Differential Reference Select Bit. Along with Bits A2–A0, this bit controls the setting of the multiplexer input, switches, and reference inputs.
1,0	PD1,PD0	Power Management Bits. These two bits decode the power-down mode of the AD7843.

Figure 3-4 shows the typical operation of serial interface of AD7843 ADC. The serial clock provides the conversion clock and controls the information transfer to and from the ADC. A complete conversion can be achieved after 24 ADC_DCLK cycles. The detailed behavior of serial port interface can be found in the datasheet of AD7843 ADC.

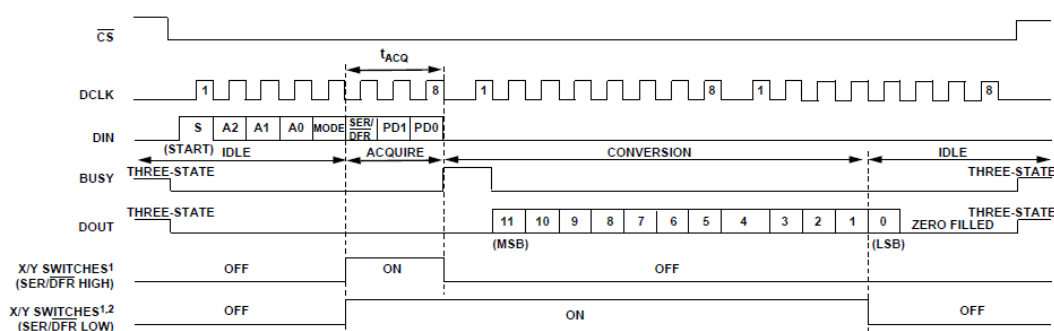


Figure 3-4 Conversion timing of serial port interface

Chapter 4

Painter Demonstration

This chapter shows how to implement the painter demonstration on LT24. The demo is designed in Qsys and running by Nios II processor. Altera SPI IP in Qsys is used to retrieve the touch information from the touch screen. Terasic custom display component in Qsys is used to display image on the 2.4" LCD. This demo requires the following hardware:

- Terasic FPGA board
- LT24 LCD touch module

4.1 Description

Figure 4-1 shows the Graphical User Interface (GUI) of Painter demo. The GUI is composed by three sections: Painting Area, Clear Button, and Color Palette. Users can select a color from the color palette and start drawing in the paint area. Click the “Clear” button to clear the painting area.

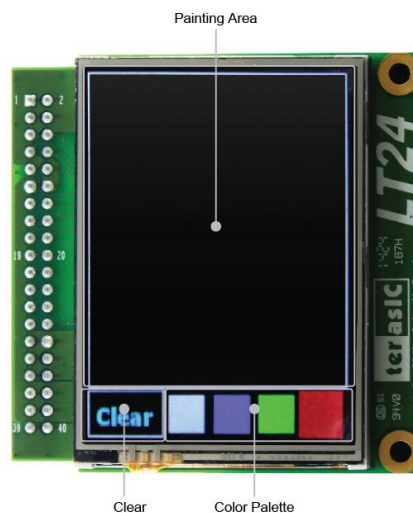


Figure 4-1 GUI of Painter demo

Figure 4-2 displays the painting on the canvas area.

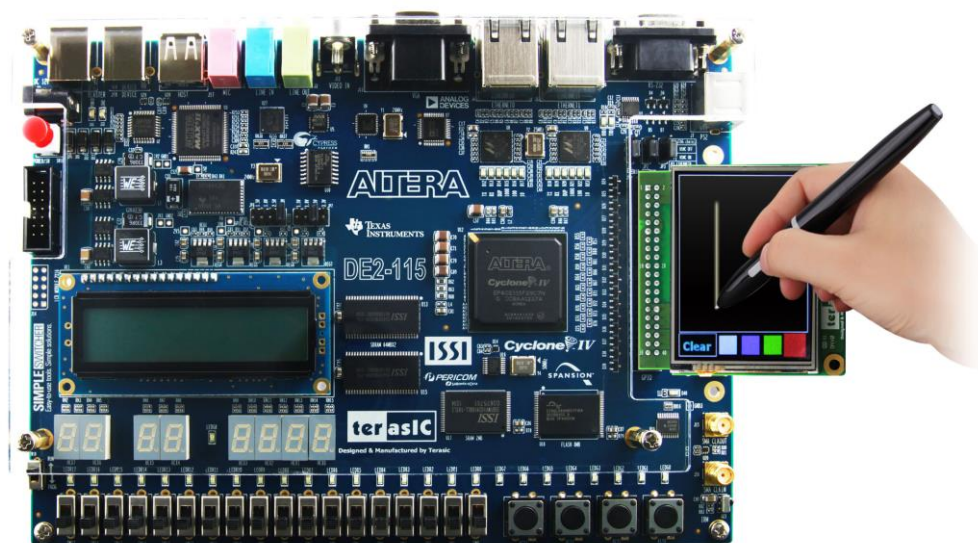


Figure 4-2 Painting demo running on LT24 with DE2-115

4.2 System Description

Figure 4-3 shows the system block diagram of Painter demonstration. Terasic custom Qsys component - **LT24 LCD controller** is used to display 240(H) x 320(V) image. Its source code is located in the “/ip/LT24_Controller” folder of Painter demo project. Qsys built-in SPI controller is used to communicate with the AD7843 ADC via SPI interface to retrieve data from the touch screen. The Nios II program handles touch event and image display. It is stored in either on-chip memory or external memory. The LCD module should be initialized before sending image data to the LCD for image display.

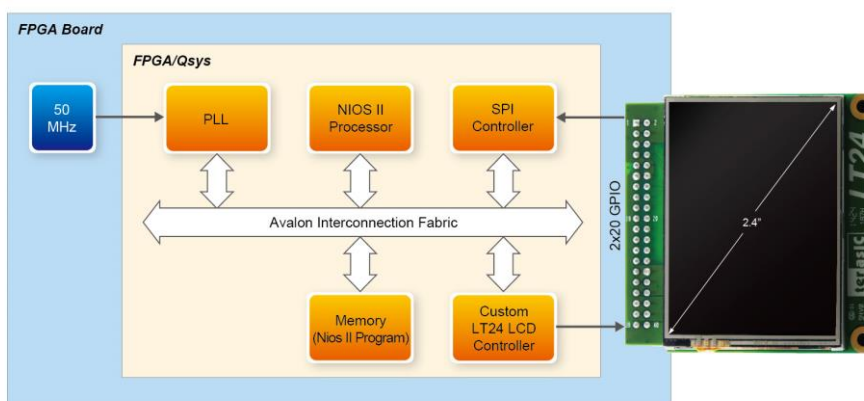


Figure 4-3 System block diagram of Painter demonstration

4.3 LT24 LCD Controller

The LT24 LCD controller is a custom Qsys component developed by Teraisc. Its source code is located in the “/ip/LT24_Controller” folder of Painter demo project. **Table 4-1** shows the register definition of LT24 LCD controller. The controller base address is defined as LT24_BASE in the Nios II program. The function of writing a control command to the LCD driver is IOWR(LT24_BASE, **0x00**, CommandValue). The function of writing data to the LCD driver is IOWR(LT24_BASE, **0x01**, DataValue).

Table 4-1 Register Definition and Offset for the Painter Demonstration

Byte Offset	Register Name	Description
0	Control Port	Write control command to the LCD driver
4	Data Port	Write data to the LCD driver

4.4 Setup Painter Demo on Terasic DE2-115 FPGA Board

This section shows how to setup the Painter demo on the Terasic DE2-115 FPGA board.

■ Hardware Setup

Figure 4-4 shows the demo setup of LT24 with DE2-115 FPGA mainboard. The LT24 should be installed on the GPIO expansion header of DE2-115.

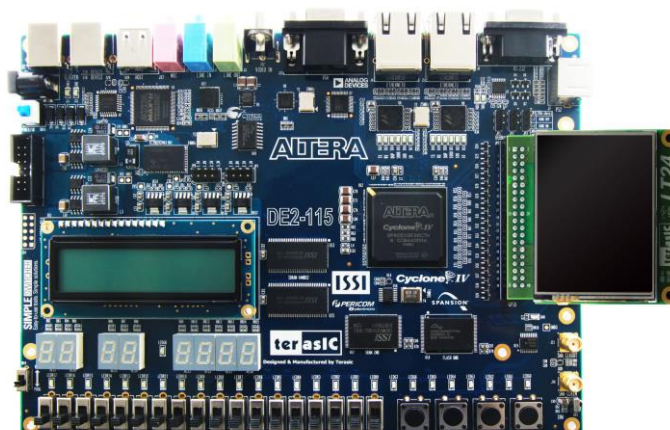


Figure 4-4 Hardware setup of LT24 with DE2-115

■ Execute Demonstration

Please follow the procedures below to setup the demonstration:

1. Power off the DE2-115.
2. Mount the LT24 onto the 2x20 GPIO 0 expansion header of DE2-115.
3. Connect the USB-Blaster USB port of DE2-115 to the USB port of host PC with a USB cable.
4. Power on the DE2-115 FPGA board.
5. Both Quartus II and USB-Blaster II driver must be installed on the host PC.
6. Copy the folder Demonstrations\DE2_115_LT24_PAINTER/demo_batch from the LT24 System CD to the host PC and execute "DE2_115.bat".
7. The Painter demo should be up and running on the LCD.

■ Source Code

The source code of Quartus project for the Painter demo with DE2-115 board is available in the "Demonstrations\DE2_115_LT24_PAINTER" folder from the LT24 System CD. The Eclipse Nios II project workspace is located in the "software" sub-folder under the Quartus project folder.

4.5 Painter Demo for Other Terasic FPGA Mainboards

The LT24 System CD also contains Quartus projects for other FPGA mainboards. The locations of source code for the Quartus projects with other Terasic FPGA mainboards are shown in [Table 4-2](#). For DE1-SoC, two Quartus projects are included in the Ssystem CD. The difference between these two projects is the memory where the Nios II program is running on. For Cyclone V GX Starter kit, there are two Quartus projects included in the LT24 System CD. The difference between these projects is also the memory where the Nios II program is running on.

To execute the demo on DE0-Nano, please connect the LT24 to the GPIO-1 expansion header of DE0-Nano. To execute the demo on DE1-SoC, please connect the LT24 to the GPIO-1 expansion header of DE1-SoC. DE0_Nano_SoC_OCRM_LT24_Painter is built by Quartus 14.1, and the other Quartus projects are built by Quartus II v13.1.

Table 4-2 Locations of Source Code for the Painter Demo with Other FPGA Boards

FPGA Board	Memory	Location
DE0-Nano	SDRAM	Demonstrations\DE0_Nano_SDRAM_LT24_PAINTER
DE2-115	On-chip	Demonstrations\DE2_115_OnChipMemory_LT24_PAINTER
DE1-SoC	SDRAM	Demonstrations\DE1_SOC_SDRAM_LT24_PAINTER
DE1-SoC	On-chip	Demonstrations\DE1_SOC_OnChipMemory_LT24_PAINTER
Cyclone V GX Starter Kit	On-chip	Demonstrations\C5G_OnChipMemory_LT24_PAINTER
Cyclone V GX Starter Kit	SRAM	Demonstrations\C5G_SRAM_LT24_PAINTER
DE0-Nano-Soc	On-Chip	DE0_Nano_SoC_OCRAM_LT24_Painter
DE2	SDRAM	DE2_SDRAM_LT24_Painter

Chapter 5

LT24_Display

This chapter shows how to implement the lt24-display demonstration on LT24. This design is only RTL code, There is not CPU, Memory. This demo requires the following hardware:

- Terasic FPGA board
- LT24 LCD touch module

5.1 Description

Figure 5-1 shows the Graphical User Interface (GUI) of LT24-Display demo. LT24-Display this demo is mainly on the LT24 can respectively show the R, G, B three color base map, as well as a simple line drawing functions. Place in the bottom three colors can click to switch between different color base map, there is a hidden feature, click on the top right corner, you can clear the lines depicted.

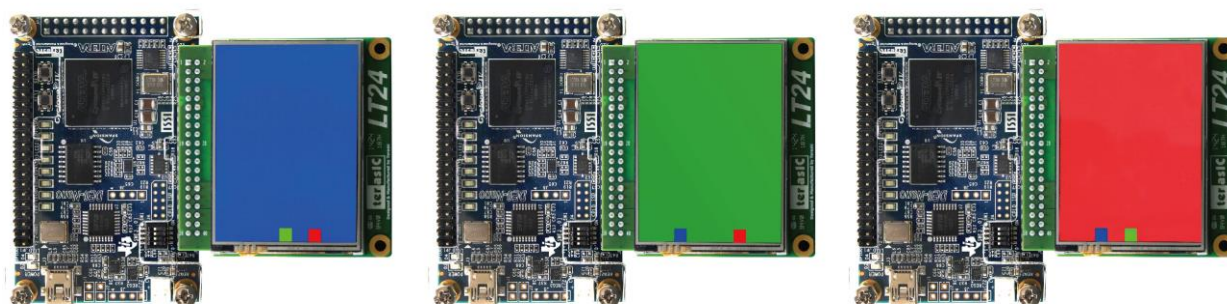


Figure 5-1 GUI of LT24-Display demo

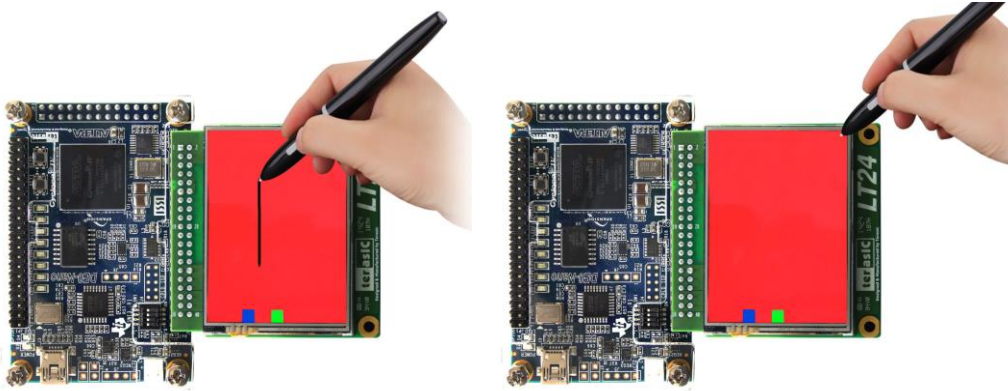


Figure 5-2 Painting demo running on LT24 with DE0-Nano

5.2 System Description

Figure 5-3 shows the system block diagram of LT24-Display demonstration. LT24_Display of RTL (Verilog) code on the demo project of "/ ip / lt24_display" folder. As Figure 5-3 shown, lt24_display the top module - lt24_display.v contains four block. Pat_update.v is used with Touch (AD7843) connection, pat_update module will receive the AD7843 in touch information (x, y coordinates), and then converted into the corresponding action, this action triggers to the LCD display module generates commands, and command again through cmd2lcd module to convert LCD panel (IL9341) can accept signals. The last timer module has only one function - to produce a 33ms signal, which is used to make a long delay of the clock, such as LCD during the initial stage will need ms delay.

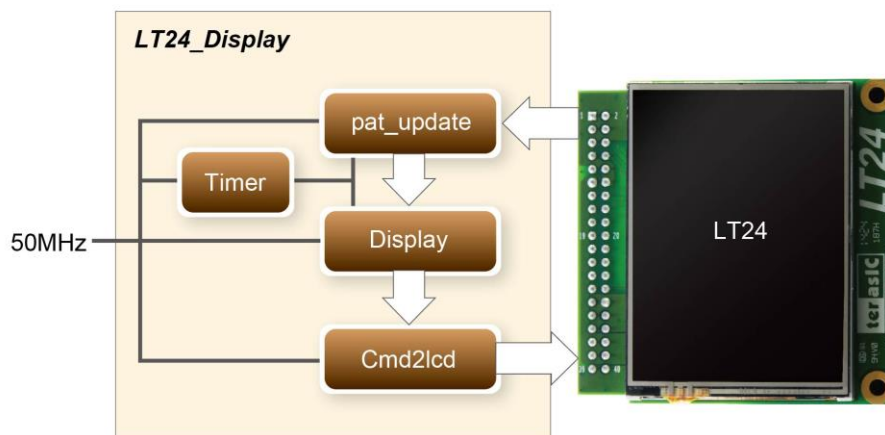


Figure 5-3 System block diagram of LT24-Display demonstration

5.3 LT24 LCD Initialization

The LCD of LT24 is needed before the initial use of. Because LT24-Display of pure RTL code design, there is no CPU, so can not be used to initial LCD mode CPU-base system (set registers to

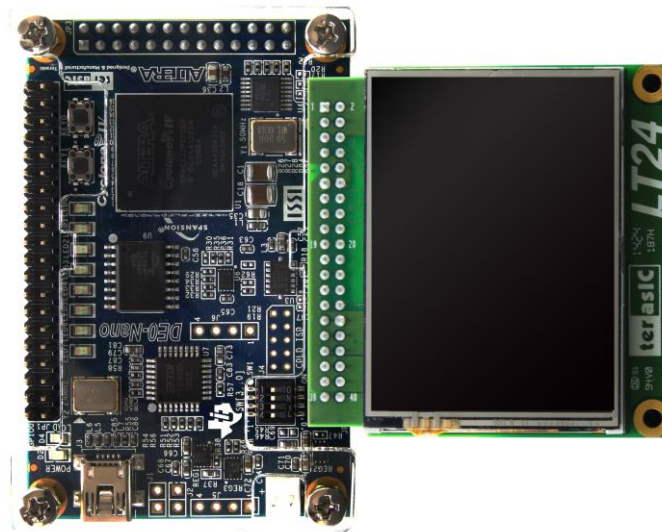
complete the process), as used herein, FSM (Final State Machine) to set the initial flow, refer to "/ip/Lt24_display/display.v" file, in initial LCD wait pat_update event after entering the loop. There CPU systems are more flexible, we have to change the process, C-code in order to change what you can; but the reaction time is slower, received from the AD7843 to output to the LCD touch response time will be longer. Compared to pure RTL code of the design, although the initial process because of the limited FSM relatively less flexible, but because the relationship between the hardware processing, AD7843 to LCD's response time faster.

5.4 Setup Demo on Terasic DE0-Nano FPGA Board

This section shows how to setup the LT24-Display demo on the Terasic DE0-Nano FPGA board.

■ Hardware Setup

錯誤! 找不到參照來源。 shows the demo setup of LT24 with DE0-Nano FPGA mainboard. The LT24 should be installed on the GPIO_1 expansion header of DE0-Nano.



■ Execute Demonstration

Please follow the procedures below to setup the demonstration:

1. Power off the DE0-Nano (disconnect USB cable).
2. Mount the LT24 onto the 2x20 GPIO 1 expansion header of DE0-Nano.
3. Connect the USB-Blaster USB port of DE0-Nano to the USB port of host PC with a USB

- cable (Power on).
4. Both Quartus II and USB-Blaster II driver must be installed on the host PC.
5. Copy the folder Demonstrations\DE0_Nano_LT24_Display\demo_batch from the LT24 System CD to the host PC and execute “test.bat”.
6. The RGB-Display demo should be up and running on the LCD.

■ Source Code

The source code of Quartus project for the LT24-display demo with DE0-Nano board is available in the “Demonstrations\DE0_nano_LT24_display” folder from the LT24 System CD.

5.5 Display Demo for Other Terasic FPGA Mainboards

The LT24 System CD also contains Quartus projects for other FPGA mainboards. The locations of source code for the Quartus projects with other Terasic FPGA mainboards are shown in Table 5-1. To execute the demo on DE0-Nano, please connect the LT24 to the GPIO-1 expansion header of DE0-Nano. To execute the demo on DE2-115, please connect the LT24 to the GPIO expansion header of DE2-115. To execute the demo on DE1-SoC, please connect the LT24 to the GPIO-1 expansion header of DE1-SoC. To execute the demo on C5G, please connect the LT24 to the GPIO expansion header of C5G. All the Quartus projects are built in Quartus II v13.1.

Table 5-1 Locations of Source Code for the LT24-Display Demo with Other FPGA Boards

FPGA Board	Location
DE0-Nano	Demonstrations\DE0_Nano_ LT24_Display
DE2-115	Demonstrations\DE2_115_ LT24_Display
DE1-SoC	Demonstrations\DE1_SOC_ LT24_Display
Cyclone V GX Starter Kit	Demonstrations\C5G_ LT24_Display

6.1 Revision History

Version	Change Log
V1.0	Initial Version (Preliminary)
V1.0.1	Add painter demo path for de0-nano-soc and DE2.

6.2 Copyright Statement

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We will continue to provide examples and lab exercises on our LT24 webpage. Please visit <http://lt24.terasic.com> for more information.